

ED303: Opening Statement

1. Learning and Teaching philosophy

Teaching practice is a foundational aspect of a pre-service teacher's career, providing essential practical experience where students can learn from experienced teachers and develop key skills "particularly in the areas of planning, teaching, classroom management, and the organisation of learning activities for pupils" (Department of Education and Science, 2006, p.1) These are all skills that I hope to develop during my teaching practice, as I transition from identifying as a university student to a pre-service teacher. Putten et al. (2014) explain that a teacher's professional identity is an important factor in determining how learning and teaching take place in a classroom. Morris & Glass (p.2) advise that teachers first begin learning to teach "by growing up in a culture – by serving as passive apprentices for 12 years or more when they themselves were students". They warn that when pre-service teachers first face the challenge of transitioning from university lectures to teaching their own class, "they often abandon new practices and revert to the teaching methods their teachers used" (Morris & Glass, p.2). Based on this, I feel it is important to clearly understand my own identity and beliefs as a teacher, by exploring how my years as a student have influenced me, and what research and theories of learning I feel are important.

My teaching philosophy and values have been heavily influenced by my time in school, with my experiences in primary and secondary school being very similar to each other. Both schools were Irish-speaking, with a predominantly middle-class student population. There was a meritocratic ethos and all teachers had a formal traditionalist approach to their teaching. The traditional teaching style is something that happened to suit me as a learner. I found that I easily understood my teachers' explanations and looked forward to spending time practicing procedural tasks and applying the latest methods they gave me. I easily understood the abstract explanations and had no difficulty interpreting my teachers' traditional style. This teaching style, however, didn't seem to fit with other students in my classes. If they couldn't understand my teachers' notes, little time was given to offer alternative explanations and many students were simply seen as not being 'maths people'. My time spent in university pursuing this BME has given me an educational context to better understand my time in school. Reflecting upon my school experience has helped me realise that much of my own learning has been dominated by a traditional 'talk and chalk' approach to mathematics, one that expected the students to replicate procedures followed by the teacher regardless of whether the student even understood what they were doing. As a Leaving Certificate student, I struggled to understand why there was an expectation that struggling students should be able to afford tutoring outside of school and spent much of my time trying to help friends in my class that were struggling. I can see that a big part of my motivation to become a teacher was in the hopes of making a difference, teaching in a way that students of all ability levels . Since my time in secondary school, the introduction of the Project Maths syllabus has highlighted the importance of a more modern approach to the teaching of mathematics. In contrast with the traditionalist style of teaching I experienced, a modern constructivist approach is centred around students' understanding and is an approach that I hope to research and utilise in my own classroom.

Constructivism is a broad term that encompasses the work of many educational psychologists, including Jean Piaget and Lev Vygotsky. While there is no single constructivist theory of learning, constructivists are said to share two main ideas: that learners play an active role in constructing their own knowledge, and that social interactions heavily influence knowledge construction (Bruning et al., 2004). In contrast with how traditionalists see learning as a passive transfer of knowledge, psychological constructivists propose that an individual's knowledge, identity and beliefs all influence how learning takes place (Woolfolk et al., 2013). Piaget (1965) advises that children advance through four stages of cognitive development, with their thinking at each stage incorporating those previous and forming a personal cognitive structure. He goes on to explain how the processes of assimilation and accommodation are used by the learner to process new information. Assimilation occurs when the learner interprets new information in the context of their current structure of cognitive development, modifying it to fit with existing knowledge and beliefs. Accommodation, on the other hand, occurs when existing cognitive structures are adapted or updated to account for the new knowledge (Piaget, 1985). A consequence of this, since learners progress through the cognitive stages at their own rate, is that learners' understanding can be limited by how they've progressed through the different developmental stages. Acknowledging that each learner constructs their own understanding at different times, Piaget's theory of Cognitive Development highlights the importance of differentiated instruction and being aware of learners' current understanding. Piaget's work has helped inform my approach to teaching. For example, I always try to revisit the prior knowledge needed for a maths lesson, as students who have had previous difficulty in understanding might benefit from a refresh. I try to ask the students what they've had difficulty with rather than assume that they learnt it previously. Learners that were previously unable to grasp certain concepts when they first encountered them might now have progressed through the stages of cognitive development and be better able to understand, contextualise and accommodate the knowledge.

Lev Vygotsky, a social constructivist, proclaimed that learning is shaped by "social interaction, cultural tools and activity" (Woolfolk et al., p.404, 2013). Through participating in activities with others, learners internalise and appropriate the knowledge and strategies of those around them. Vygotsky (1962) believes that learning takes place inside the 'Zone of Proximal Development'. In this zone, learners can master concepts they wouldn't fully grasp on their own by working together with others that are more advanced. Learning in this zone takes place through intersubjectivity and scaffolding. Intersubjectivity defines the process that occurs when two learners who begin a task with different levels of understanding grow towards a shared understanding through communication and adjusting to each other's perspectives. Scaffolding, on the other hand, involves the teacher adjusting the level of their help to best fit the current understanding of the learner and maximise their learning (Woolfolk et al., 2013).

Both Piaget and Vygotsky highlight the importance of viewing the teacher as a facilitator of a student's active construction of knowledge, rather than the traditionalist practice I experienced in school. In my opinion, in order to be an effective teacher an effort should be made to account for the diversity of any given classroom. Content should be adapted so that it is taught in the context of real life applications, and lessons should be designed in a way to promote student discussion and foster their natural curiosity for learning. After outlining the context of the school where I will be undertaking placement in Section 2, I will highlight how these theories of learning have informed the learning goals of my teaching practice and the methodologies I hope to incorporate.

2. School Context

My teaching practice will take place in a local post-primary school in Galway City that facilitates 300 students. These students are of diverse socio-economic and cultural backgrounds, and the school prides itself in promoting both academic and vocational achievement. I will have 2 roles in the school, one as a mainstream teacher teaching a single class, and another as a resource teacher during after-school study.

My mainstream class is with the school's Transition Year students, where the class is a mixture of higher level and ordinary level students. The class has 22 students, attendance fluctuates wildly, however. Among these students are 4 English as an Additional Language students. In communication with my co-operating teacher, the maths topic of my classes will be decided weekly based on what the co-operating teacher is currently teaching. In terms of resources, the students don't have a textbook assigned to them for Transition Year and lessons will have to be designed around the use of preparing materials in advance as well as utilising the whiteboard and projector available in the classroom.

My role as a resource teacher will involve spending two hours per week supervising the school's Evening Study. This is a service provided free of charge and supports students from 1st Year to 6th Year by providing them a comfortable place complete their homework or study. I hope to use this time to support students with their homework or study, and I see it as an opportunity to help improve students' learning in other classes by helping them with any work they're currently struggling with.

3. Learning Goals for Placement

I view this teaching practice as a solid foundation for my future career as a teacher. In this sense, it's important to me that my teaching is informed by research. I have previously outlined educational theory that I feel has improved my understanding of student learning, and I hope to design my lessons in a way that facilitates this. My goals are to:

- Utilise problem-based learning, to promote discussion in class and move away from a teacher-centred approach.
- Identify differentiated learning goals to make sure each lesson is designed in a way to support all students.
- Make use of mixed-ability grouping, to promote peer learning through intersubjectivity.
- Improve my questioning skills and integrate scaffolding into my practice.

It's important to plan different actions throughout my placement to ensure I'm working towards these goals:

- Lesson Planning. Without a lot of time spent planning lessons, I won't be able to create the type of learning environment that I've set out in my goals above. I will make use of high-quality resources, as well as skills developed in other modules such as 'Mathematics Methodologies & Skills of Teaching', to try to design lessons as best I can.
- Reflective Practice. As a pre-service teacher I believe that I have a lot to learn, and in order for me to benefit from research-informed practice I need to be able to see how my efforts translate to the classroom. By undertaking weekly reflections, I hope to learn from any successes or problems that might arise. This will be a consistent way for me to understand why different ideas did or didn't work and help me improve my practice for the future.
- Classroom Observations. My co-operating teacher has advised that I can observe her other classes when our timetables allow, which is something that I will be sure to do. Learning from a more experienced teacher and seeing how they manage classrooms and design lessons will be an invaluable resource during placement.

Bibliography

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